

T E C H R E F E R E N C E

Feature Engineering Technical Reference

11 Disciplines · Economics · Chemistry · Epidemiology · Criminology · Wave · TDA ·
HMM · GMM · Mamba · Graph · 3-Stage Normalization

Project

AWS PLE for Financial

PLE-adaTT Multi-Task Recommendation

Version

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This document describes the complete feature engineering pipeline of the AWS PLE for Financial project. It covers the theoretical foundations, mathematical definitions, output feature specifications of 11 cross-disciplinary feature generators (Economics, Chemical Kinetics, SIR, Crime Pattern, Wave Interference, TDA, HMM, GMM, Mamba, Graph, Base), and the 3-Stage Normalization Pipeline.

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Design vs. Implementation Dimensions

This document is written based on the **full-bank design (734D)**. The current Santander benchmark implementation uses **1211D input (17 feature groups)**. For the actual implementation dimension specifications, refer to `outputs/phase0/feature_schema.json`. The Appendix “Design vs. Implementation Dimension Mapping” details the per-group differences.

Feature Design Philosophy

Rationale for a Multidisciplinary Approach

Traditional feature engineering views data through a single lens: statistics. Means, variances, frequencies, and correlations are powerful, yet they reveal only a fraction of the structure inherent in the data. This system simultaneously applies the mathematical frameworks of 11 academic disciplines to extract different facets of the data.

Discipline	Patterns Captured	What Statistics Alone Cannot See
Economics	Income structure, elasticity, time preference	Permanent vs. transitory income decomposition, causal consumption direction
Chemical Kinetics	Speed, barriers, and catalysts of transformation	Energy barriers of state transitions, second-order derivatives (acceleration)
Epidemiology (SIR)	Adoption diffusion, immunity/churn	Population-level dynamics, transmission threshold R_0
Criminology	Routines, deviations, anomalous patterns	Circular nature of time, Burstiness
Wave Physics	Periodic decomposition, phase synchronization	FFT frequency domain, PLV phase coupling
TDA	Topological shape, holes, connectivity	Coordinate-invariant structure, multi-resolution observation
HMM	Hidden state transitions	Probabilistic inference of unobservable latent stages
GMM	Soft clustering	Probabilistic type assignment, uncertainty quantification
Mamba SSM	Long-range temporal dependencies in time series	Nonlinear temporal representation via selective state spaces
Graph (LightGCN)	Collaborative filtering signals	Multi-hop indirect preferences, hierarchical structure
Base Statistics	Demographics, holdings, transactions	RFM, category distribution, channel diversity

Structural Isomorphism

In multidisciplinary feature engineering, “analogy” is not mere metaphor — it is grounded in **structural isomorphism**. Two systems may differ in their surface objects (molecules vs. consumers), yet when the *relational structure* among those objects is mathematically identical, a structural isomorphism holds.

Core Principle

The “time for reactant concentration to halve” in chemistry and the “time for transaction frequency to halve” in finance both share the same mathematical structure: exponential decay. When the equations are identical, the patterns they capture are identical. What the objects represent does not affect the validity of the equations.

The Dual Role of Features

Features in this system serve two simultaneous roles.

1. **Predictive input:** As input tensors to the PLE-adaTT model, they contribute to the prediction of 12 tasks.
2. **Expert routing signal:** The soft assignment probabilities (γ_{nk}) from GMM are used as routing signals that weighted-combine the 20 sub-heads of the GroupTaskExpertBasket, and the 48D HMM output is fed to a dedicated Projector through a separate input path.

Expert Routing Granularity

Expert routing (`target_experts` in `feature_groups.yaml`) is declared at the **feature-group level**, not the column level. Each expert receives the full subset of its assigned feature groups as sliced by `FeatureRouter` . Column-level routing is not supported and would break the config-driven design.

Overall Feature Tensor Composition

Feature Block	Dim.	Components
Base	238D	RFM(34) + Category(64) + Transaction Stats(76) + Product Diversity(12) + ...
Multi-Source	91D	Deposit + Credit + Investment + Digital, etc.
Extended-Source	84D	Insurance + Refund + Consultation + STT, etc.
Domain	159D	TDA(70) + GMM(22) + Mamba(50) + Economics(17)
Model-Derived	27D	HMM summary(5) + Bandit/MAB(4) + LNN(18)
Multidisciplinary	24D	Chemical(6) + SIR(5) + Crime(5) + Wave(8)
Merchant Hierarchy	27D	MCC Poincaré embeddings (hierarchy) + brand embeddings (was 21D stats only)
Total (normalized)	644D	
Raw power-law copy	90D	log1p originals of power-law columns (unscaled)
Main Tensor Total	734D	644D normalized + 90D raw power-law

Separate input: HMM Triple-Mode 48D + Hyperbolic 20D = 68D (separate input path)

Economics Features (Economics, 17D)

Domain Features

Friedman PIH

17D / 734D

Theoretical Foundation: Friedman’s Permanent Income Hypothesis

Friedman’s (1957) Permanent Income Hypothesis (PIH) decomposes observed income into a permanent component and a transitory component.

$$Y_t = Y_t^P + Y_t^T$$

Here Y_t^P is permanent income (long-run stable) and Y_t^T is transitory income (temporary fluctuations). The consumption function follows:

$$C_t = k(r, w, u) \cdot Y_t^P$$

Core implication: consumers spend in proportion to their permanent income; transitory income flows into savings and investment.

Income Decomposition Estimation Methods

Method	Formula	Complexity
Moving Average	$\hat{Y}_t^P = \frac{1}{L} \sum_{i=0}^{L-1} Y_{t-i}, L = 12$	Low
HP Filter	$\min_{\tau} \left\{ \sum_t (Y_t - \tau_t)^2 + \lambda \sum_t [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2 \right\}$	Medium
Kalman Filter	State: $Y_{t+1}^P = Y_t^P + \eta_t$, Obs: $Y_t = Y_t^P + \varepsilon_t$	High

The first-order condition of the HP Filter yields $(I + \lambda D^\top D)\tau = Y$, a positive-definite banded linear system solvable in $O(T)$ via Cholesky decomposition. Following the Ravn-Uhlig (2002) standard, $\lambda = 14400$ is used for monthly data.

Income Decomposition Output (8D)

Feature	Formula	Financial Interpretation
permanent_income_avg	$\text{mean}(\hat{Y}^P)$	Long-run stable income level
permanent_income_stability	$\frac{\sigma(\hat{Y}^P)}{\mu(\hat{Y}^P)}$	CV of permanent income; low value implies stable employment
permanent_income_growth	$\frac{\hat{Y}_T^P - \hat{Y}_1^P}{\hat{Y}_1^P}$	Income trajectory (indicator for card tier upgrade)

Feature	Formula	Financial Interpretation
<code>permanent_income_trend</code>	REGR_SLOPE / polyfit	Robust long-term growth direction
<code>transitory_income_avg</code>	$\text{mean}(\hat{Y}^T)$	Bonus frequency indicator (theoretically ≈ 0)
<code>transitory_income_volatility</code>	$\sigma(\hat{Y}^T)$	Magnitude of income uncertainty
<code>transitory_income_max</code>	$\max(\hat{Y}^T)$	Proxy for the largest bonus event
<code>bonus_frequency</code>	$\frac{\text{count}(\hat{Y}^T > 0.5 \hat{Y}^P)}{N}$	Proportion of periods with a large bonus

Microeconomic Behavioral Features (9D)

Income Elasticity

$$\varepsilon_Y = \frac{\partial Q}{\partial Y} \cdot \frac{Y}{Q} = \frac{d \ln Q}{d \ln Y}$$

- $\varepsilon_Y > 1$: propensity for luxury consumption (spending grows faster than income)
- $0 < \varepsilon_Y < 1$: propensity for necessity consumption
- $\varepsilon_Y < 0$: inferior good (consumption declines as income rises)

Implemented as the discrete arc elasticity: $\hat{\varepsilon}_Y = \frac{1}{T} \sum_{t=1}^T \frac{\Delta \frac{S_t}{S_{t-1}}}{\Delta \frac{Y_t}{Y_{t-1}}}$

Consumption Smoothing

Hall (1978) combined the PIH with rational expectations to show that optimal consumption follows a random walk.

$$C_t = C_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

`consumption_smoothing` = $\frac{\mu}{\sigma}$ (inverse CV, the Sharpe ratio of consumption). Higher values indicate closer proximity to the theoretical optimum.

Time Discounting

$$V_0 = \sum_{t=0}^T \beta^t u(C_t), \quad 0 < \beta < 1$$

`discount_rate_proxy` = first-half spending / second-half spending. A larger value implies a smaller $\beta \eta$ and stronger present bias — useful for identifying customers who prefer immediate rewards (same-day cashback).

Other Behavioral Features

Feature	Formula	Interpretation
spending_diversification	$H = -\sum_i s_i \ln(s_i)$	Shannon entropy; category diversity
category_hhi	$\text{HHI} = \sum_i s_i^2$	Herfindahl index; spending concentration
savings_propensity	$\frac{Y-C}{Y}$	Savings propensity; reflects expected future utility
price_sensitivity	Refund ratio proxy	Behavioral proxy for price elasticity
cross_category_elasticity	Time variation in number of categories	Cross-elasticity proxy

Shannon entropy and HHI are special cases of Rényi entropy $H_\alpha = \frac{1}{1-\alpha} \ln(\sum s_i^\alpha)$: Shannon corresponds to $\alpha \rightarrow 1$, and HHI to $\alpha = 2$ ($H_2 = -\ln(\text{HHI})$).

Chemical Kinetics Features (Chemical Kinetics, 6D)

Multidisciplinary Arrhenius Equation 6D / 24D

Theoretical Foundation

In the physicochemical worldview, every transformation (reaction) occurs on an energy landscape. For a system to move from one state to another, it must surmount an energy barrier; the height of that barrier determines the difficulty of the transition, while the frequency of surmounting it determines the rate.

Arrhenius Equation

$$k = Ae^{-E_a/RT}$$

The category-switching frequency (k) increases exponentially when the entry barrier (E_a) decreases or consumer activity (T) increases.

Half-Life

The half-life of a first-order reaction, $T_{1/2} = \ln 2/k$, corresponds to the median transaction interval. Customers with a short half-life exhibit a rapid transaction cycle.

Second-Order Finite Difference

Spending acceleration is the discrete second-order derivative:

$$f''(t) \approx f(t + \Delta t) - 2f(t) + f(t - \Delta t)$$

Derived by cancellation of the f' term in the Taylor expansion, it distinguishes acceleration (positive) from deceleration (negative).

Output Features (6D)

Feature	Definition	Financial Interpretation
<code>new_category_activation_rate</code>	New MCCs in last 30 days / active MCCs	Proxy for inverse activation energy
<code>spending_half_life</code>	Median transaction interval (days)	Chemical half-life $T_{1/2} = \ln \frac{2}{k}$
<code>spending_acceleration</code>	$f(t + 1) - 2f(t) + f(t - 1)$	Second-order finite difference: acceleration (+) or deceleration (-)
<code>dormancy_reactivation_rate</code>	MCC present in W1, absent in W2, reappearing in W3	Catalytic reactivation rate

Feature	Definition	Financial Interpretation
catalyst_sensitivity	Daily avg. spend in early month / daily avg. spend in late month	Payday catalyst elasticity
saturation_proximity	$\frac{\text{max}}{\text{avg} + \text{std}}$	Proximity to spending ceiling

SIR Epidemiology Features (Epidemiology, 5D)

Multidisciplinary

Compartmental Model

5D / 24D

Theoretical Foundation: Kermack-McKendrick Model

$$\frac{dS}{dt} = -\beta SI, \quad \frac{dI}{dt} = \beta SI - \gamma I, \quad \frac{dR}{dt} = \gamma I$$

Basic reproduction number: $R_0 = \frac{\beta}{\gamma}$. When $R_0 > 1$, adoption spreads in a self-reinforcing manner; when $R_0 < 1$, it naturally dies out.

Financial Domain Mapping

Compartment	Epidemiology	Financial Interpretation
S (Susceptible)	Uninfected	Unused categories among the top-15 MCC by population
I (Infected)	Currently spreading	Daily avg. frequency in last 30 days > prior period
R (Recovered)	Immune/recovered	Categories used in the past but inactive in last 30 days

The infection classification criterion applies a period-length correction factor of $\frac{30}{L-30}$.

Output Features (5D)

Feature	Description
<code>sir_susceptible</code>	S ratio. High value indicates a wide recommendation opportunity space (exploration-ready)
<code>sir_infected</code>	I ratio. High value indicates active adoption underway (optimal timing for cross-sell)
<code>sir_recovered</code>	R ratio. High value indicates contraction (target for retention campaigns)
<code>sir_r0</code>	Basic reproduction number. Degree of self-reinforcing adoption spread
<code>sir_infection_rate</code>	Adoption speed. Corresponds to βS

Customer Profiling

Profile	S	I	R	Interpretation
Exploration-Ready	High	Low	Low	Maximum recommendation opportunity space
Active Adoption	Mid	High	Low	Optimal moment for cross-selling
Stable Usage	Low	Low	Low	Loyal customer
Contraction	Low	Low	High	Target for churn-prevention campaign

Criminology Features (Crime Pattern / Routine Activity, 5D)

Multidisciplinary Routine Activity Theory 5D / 24D

Theoretical Foundation

In Cohen & Felson's (1979) Routine Activity Theory, behavior is governed by daily routines, and deviations from those routines create opportunities for anomalous events.

Burstiness (Barabasi, 2005)

$$B = \frac{\sigma_\tau - \mu_\tau}{\sigma_\tau + \mu_\tau} \in [-1, 1]$$

- $B = -1$: perfectly regular intervals (recurring payments)
- $B = 0$: Poisson process (random)
- $B = +1$: extreme clustering (explosive shopping)

Human behavior follows not a Poisson process but a heavy-tailed distribution characterized by short bursts followed by long rests.

Circular Statistics

Essential for analyzing transaction timestamps. The distance between 11 PM and 1 AM is 2 hours, not 22.

$$\overline{R} = \left(\frac{1}{n} \sum \cos \theta_i, \frac{1}{n} \sum \sin \theta_i \right), \quad CV = 1 - |\overline{R}|$$

where $\theta = 2\pi h/24$. Euclidean distance distorts circular time, whereas circular statistics correctly measures concentration.

Output Features (5D)

Feature	Description
<code>txn_burstiness</code>	Burstiness of transaction intervals. Regularity vs. explosiveness of spending rhythm
<code>time_circular_variance</code>	Circular variance of transaction timestamps. Concentration across time-of-day
<code>routine_stability</code>	Routine stability. Consistency of day-of-week patterns

Feature	Description
<code>breakpoint_count</code>	Number of breakpoints. How many times the spending routine was disrupted
<code>anomaly_score</code>	Anomaly pattern score. Intensity of deviation from routine

Wave Interference Features (Wave Interference, 8D)

MultidisciplinarySpectral Analysis8D / 24D

Theoretical Foundation

In the wave-physics worldview, when multiple waves superpose, their phase relationship produces either constructive interference (amplitude increase) or destructive interference (amplitude decrease).

KL Divergence

$$D_{\text{KL}(P\|Q)} = \sum P(x) \ln \frac{P(x)}{Q(x)}$$

The KL divergence between weekday and weekend spending distributions quantifies, in *bits*, how differently a customer behaves across the two contexts. By Gibbs’ inequality (a consequence of Jensen’s inequality), it is always non-negative.

Phase Locking Value (PLV)

A concept borrowed from functional connectivity in neuroscience.

$$\text{PLV} = \frac{1}{T} \left| \sum_{t=1}^T e^{j(\varphi_x(t) - \varphi_y(t))} \right|$$

It measures the consistency of the phase difference between the spending rhythms of two categories, extracted via the Hilbert transform.

Cross-spectral Coherence

$$C_{xy}(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f) \cdot S_{yy}(f)}$$

A frequency-resolved correlation that identifies at which periodicities categories co-synchronize.

Output Features (8D)

Feature	Description
weekday_weekend_kl	KL divergence between weekday and weekend spending distributions
spectral_entropy	Normalized Shannon entropy of the power spectrum. Predictability of spending periodicity

Feature	Description
<code>dominant_period</code>	Period (days) of the strongest frequency component identified by FFT
<code>phase_locking_value</code>	PLV. Phase synchronization of spending rhythms across categories
<code>cross_coherence</code>	Cross-spectral coherence. Per-frequency category correlation
<code>hhi_shift</code>	Change in HHI. Trend toward spending concentration or diversification
<code>category_sync_ratio</code>	Synchronization ratio. Proportion of category pairs in phase
<code>interference_strength</code>	Interference strength. Degree of constructive/destructive interference across categories

Information-Theoretic Justification

The four multidisciplinary modules capture nearly orthogonal projections:

- **Chemical Kinetics:** the *differential structure* of time (first- and second-order derivatives)
- **Epidemiological Diffusion:** the *state-space transition structure* (S -> I -> R)
- **Crime Patterns:** the *statistical texture* of time series (periodicity, clustering, dispersion)
- **Wave Interference:** the *frequency-domain spectral structure* (FFT, coherence, phase)

Cross-module combinations reveal patterns invisible to individual modules: for example, high `catalyst_sensitivity` + high `burstiness` = payday-burst spender (optimal target for early-month promotional campaigns).

TDA Features (Topological Data Analysis, 70D)

Domain Features Persistent Homology 70D / 159D

Core Idea

Core Principle of TDA

By growing balls of increasing radius around data points, we track *when* topological characteristics such as connected components, holes, and voids *appear and disappear*. Long-lived characteristics are signal; short-lived ones are noise.

Strengths of TDA:

1. **Coordinate invariance:** topological characteristics are preserved under rotation and continuous deformation of the data. Robust to feature scaling.
2. **Multi-resolution observation:** all scales are considered simultaneously, rather than a single threshold.
3. **Noise resistance:** by retaining only long-lived characteristics, spurious structure is naturally filtered out.

Betti Numbers and Homology

$$\beta_k = \text{rank}(H_k(X))$$

$H_k(X)$ is the k -dimensional homology group of space X .

- β_0 : number of connected components (how many clusters the points form)
- β_1 : number of one-dimensional holes (how many loops/cycles exist)
- β_2 : number of two-dimensional voids (how many enclosed cavities exist)

Homology identifies chains that have no boundary and are not themselves the boundary of anything else:

$$H_k = \text{Ker}(\partial_k) / \text{Im}(\partial_{k+1})$$

Financial Domain Interpretation

- **Large H_0 :** spending falls into several disconnected clusters (living expenses vs. travel vs. education)
- **Non-trivial H_1 :** periodic circular pattern (month-start card payment → mid-month food → month-end leisure → repeat)
- **Non-trivial H_2 :** “hollow” pattern in 3D structure (central void in the amount–category–time space)

Vietoris-Rips Complex and Persistent Homology

This system employs the Vietoris-Rips complex:

$$\sigma = \{x_0, \dots, x_k\} \in \text{VR}_\varepsilon(X) \quad \text{iff} \quad d(x_i, x_j) \leq \varepsilon, \forall i, j$$

It is a “2× radius” approximation of the Čech complex, but computationally far more efficient (requires only pairwise distance comparisons).

Persistent Homology observes *all scales simultaneously*, resolving the sensitivity to a single threshold. Points far from the diagonal in a Persistence Diagram represent strong signals.

Stability Theorem

$$d_B(\text{Dgm}(f), \text{Dgm}(g)) \leq \|f - g\|_\infty$$

Perturbations to the input are bounded by the maximum change in the filtration function. This mathematically guarantees the noise robustness of topological features.

Output Features (70D)

Subgroup	Dim.	Description
tda_short	24D	Short-term topological patterns from 90-day app logs
tda_long	36D	Long-term topological patterns from 12-month financial transactions
phase_transition	10D	Topological change detection across time windows

Relationship to the PersLay Expert

The TDA features in this document (offline 70D) and the PersLay Expert (online 64D) both leverage Persistent Homology but serve different roles. The 70D features are part of the 734D main tensor and are precomputed during batch preprocessing. The PersLay Expert learns end-to-end from Persistence Diagrams inside the PLE model.

HMM Features (Hidden Markov Model, 5D summary + 48D separate)

Model-Derived + Separate Input Triple-Mode HMM 53D total

Core Idea: Hidden State Inference

What can be directly observed from card transaction data is limited to transaction amount, frequency, and category diversity. Yet even an identical 100,000-won monthly spend may belong to two entirely different states: “a customer exploring a new service” versus “a last-gasp spend before churn.” The HMM **probabilistically infers** unobservable latent states from observed data.

Markov Property

$$P(q_{t+1} \mid q_t, q_{t-1}, \dots, q_1) = P(q_{t+1} \mid q_t)$$

The next state depends only on the current state. This simplification bounds the parameter count to $O(N^2)$, enabling stable learning across hundreds of thousands of customers.

Triple-Mode Architecture

Mode	States	Output	Patterns Captured
Journey	5	16D	Daily/weekly customer journey (AWARENESS → CONSIDERATION → PURCHASE → ...)
Lifecycle	5	16D	Monthly/yearly lifecycle (NEW → GROWING → MATURE → AT_RISK → ...)
Behavior	6	16D	Monthly behavioral pattern types (frugal, investment-oriented, consumption-oriented, etc.)

The 48D output from the three modes is supplied as a separate input to the PLE’s HMM Triple-Mode Projector.

Three Core Algorithms

Problem	Question	Algorithm	Usage
Evaluation	What is the likelihood of a sequence?	Forward	Model quality validation, anomaly detection
Decoding	What is the optimal state sequence?	Viterbi	State sequence extraction → meta features

Problem	Question	Algorithm	Usage
Learning	What are the optimal parameters?	Baum-Welch (EM)	Optimize transition/emission distributions

Forward-Backward

$$\alpha_t(i) = P(o_1, \dots, o_t, q_t = S_i \mid \lambda), \quad \gamma_t(i) = P(q_t = S_i \mid \mathbf{O}, \lambda)$$

$\gamma_{t(i)}$ is the soft state-assignment probability and constitutes the core of the features. A probability vector such as “80% active, 15% growing, 5% at-risk” is itself a rich feature.

HMM Summary (Main Tensor 5D)

Feature	Description
<code>hmm_dominant_state</code>	State ID with the highest $\gamma_{t(i)}$ aggregated across the three modes
<code>hmm_state_duration</code>	Consecutive duration (months) spent in the current dominant state
<code>hmm_transition_stability</code>	Inverse of the state-transition frequency over the last N months
<code>hmm_transition_entropy</code>	$H = -\sum_j a_{ij} \log a_{ij}$. High value implies difficulty predicting the next state
<code>hmm_state_change_rate</code>	Number of state changes / total observation period

HMM vs. GMM

HMM performs *time-series* state inference: “What *stage* is this customer currently in, and where will they move next?” GMM performs *cross-sectional* clustering: “What *type* is this customer?” Because the transition matrix $\mathbf{A}$ captures temporal dynamics, the HMM can distinguish “currently growing” from “on the verge of churn” even for identical spending patterns. The two modules are supplied to the PLE through separate input paths and are mutually complementary.

GMM Features (Gaussian Mixture Model, 22D)

Domain Features EM Algorithm + BIC 22D / 159D

Theoretical Foundation

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

where $\pi_k \geq 0$, $\sum_k \pi_k = 1$, and each component is a multivariate Gaussian. Architecture: $K = 20$ clusters, $D = 40$ input dimensions, `covariance_type = "full"`.

Core assumption: “The observed data were generated from K subpopulations, and each data point’s group membership is unobserved (a latent variable).”

EM Algorithm

E-Step (Posterior Responsibility)

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

A direct application of Bayes’ theorem: the prior π_k and likelihood $\mathcal{N}_k$ are combined to yield the posterior γ_{nk} .

M-Step

γ_{nk} is used as a weight to update $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, and π_k .

Convergence Guarantee

By Jensen’s inequality, EM monotonically non-decreases the ELBO. Global optimality is not guaranteed; this is mitigated with `n_init=10`.

Model Selection: BIC

$$\text{BIC} = -2 \ln \hat{L} + k \ln(n)$$

With hundreds of thousands of data points, BIC’s $k \ln(n)$ penalty prevents overfitting more effectively than AIC’s $-2 \ln \hat{L} + 2k$.

Output Features (22D)

Feature	Dim.	Description
cluster_prob_00 – 19	20D	Soft assignment probabilities γ_{nk} (sum = 1.0)
cluster_id	1D	Hard assignment $\arg \max_k \gamma_{nk}$
cluster_entropy	1D	$H_n = -\sum_k \gamma_{nk} \ln(\gamma_{nk} + \varepsilon)$

Entropy interpretation:

- $H \approx 0$: a clear behavioral archetype (high-confidence classification)
- $H = \ln(20) \approx 2.996$: uniform distribution (cold-start / unclassifiable)

GMM vs. K-Means: Why Soft Assignment?

Perspective	K-Means	GMM
Assignment	Hard: one-hot (1 bit)	Soft: probability vector (4.32 bits max)
Boundary customers	Arbitrary assignment, unstable	Probability spread across adjacent clusters
Cluster shape	Spherical (Euclidean)	Ellipsoidal (Mahalanobis, full covariance)
Uncertainty	None	Entropy-based confidence
PLE role	Single sub-head activation	γ_{nk} -weighted ensemble combining 20 sub-heads

The Mahalanobis distance $d_M = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$ generalizes Euclidean distance ($\boldsymbol{\Sigma} = \mathbf{I}$) by accounting for feature correlations and scale differences.

Mamba Time-Series Features (Selective State Space, 50D)

Domain Features

Mamba SSM + PCA

50D / 159D

Core Idea

Simple aggregate statistics (sum, mean, standard deviation) completely destroy the **temporal ordering** of a time series. Shuffle the values randomly and the sum and mean remain unchanged. Mamba SSM captures nonlinear long-range dependencies in time series via a selective state-space mechanism.

Selective State Space Model

Mamba is based on the discretization of the Continuous State Space Model (SSM):

Continuous: $h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t)$

Discretization: $\overline{A} = \exp(\Delta A), \quad \overline{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B$

Key point: Because Δ_k varies **dynamically** with the input, the model adaptively updates its state even for non-stationary time series.

Advantages over Prior Methods

Perspective	RNN	Transformer	Mamba
Long-range dependency	Vanishing gradient	Resolved via self-attention	Resolved via selective SSM
Sequence length	$O(T)$ sequential	$O(T^2)$ memory	$O(T)$ linear
Non-stationarity	Difficult to adapt	Dependent on positional encoding	Δ_k dynamic adaptation

Pipeline

- MCC 15D × 180-day input sequence
- Mamba encoder (`d_model=256`) extracts hidden representations
- PCA → 50D compression
- Integrated into the Domain block of the 734D main tensor

Distinction from the Online Mamba Expert

These features (offline 50D) use `d_model=256` with a 15D MCC $\times$ 180-day input and are precomputed *before* PLE training. The Temporal Expert’s Mamba inside PLE uses `d_model=128` with a 16D card $\times$ 180-step input and is trained end-to-end. No weight sharing.

Four Perspectives of Time-Series Analysis

Mamba 50D is a *learned representation* that nonlinearly integrates information from the following four perspectives:

Perspective	Key Question	Technique
Time domain	How do values change over time?	Autocorrelation, change points, moving average
Frequency domain	What periodicities repeat?	FFT, spectral analysis
Distribution/shape	What shape does the value distribution take?	Skewness, kurtosis, tail probability
Information theory	How complex and predictable is the series?	Entropy, permutation entropy

Graph Features (LightGCN + Hyperbolic GCN)

Offline Precomputed Graph Neural Network 64D + H-GCN aggregation

LightGCN: Collaborative Filtering

Message Passing

$$e_u^{(k+1)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u|} \cdot \sqrt{|\mathcal{N}_i|}} \cdot e_i^{(k)}$$

Symmetric normalization $\tilde{A} = D^{-1/2} A D^{-1/2}$ simultaneously suppresses the influence of popular items (sender) and high-degree receivers.

Layer Combination

$$e_u^{\text{final}} = \frac{1}{L+1} \sum_{k=0}^L e_u^{(k)}$$

Uniform average of the 0-hop (self) through 1-, 2-, 3-hop neighbors. Empirically superior without learnable attention weights, and prevents overfitting.

BPR Loss

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u, i^+, i^-)} \log \sigma(\hat{y}_{ui^+} - \hat{y}_{ui^-}) + \lambda \|\Theta\|^2$$

A pairwise ranking loss that optimizes relative ranking rather than absolute scores.

Hyperbolic GCN: Hierarchical Structure

Poincaré Ball Model

$$\mathbb{B}_c^d = \{\mathbf{x} \in \mathbb{R}^d : c\|\mathbf{x}\|^2 < 1\}$$

Exponential map: $\exp_{\mathbf{0}}(\mathbf{v}) = \tanh(\sqrt{c}\|\mathbf{v}\|) \cdot \frac{\mathbf{v}}{\sqrt{c}\|\mathbf{v}\|}$

Poincaré distance: $d_{\mathbb{B}}(\mathbf{x}, \mathbf{y}) = \frac{1}{\sqrt{c}} \operatorname{arccosh}\left(1 + \frac{2c\|\mathbf{x}-\mathbf{y}\|^2}{(1-c\|\mathbf{x}\|^2)(1-c\|\mathbf{y}\|^2)}\right)$

Near the boundary the denominator approaches 0, so the hyperbolic distance explodes even when the Euclidean distance is small. This is the mechanism that naturally encodes hierarchical depth.

Financial Domain Justification

The MCC taxonomy (Root → L1(8) → L2(100) → Brand(50K) → Branch(500K)) is inherently a tree structure. Hyperbolic space matches tree branching through exponential volume growth, embedding 550K nodes in an 8D Poincaré Ball. Nickel & Kiela (2017) result: 5D hyperbolic > 200D Euclidean (WordNet hierarchy).

Dual GCN Architecture

Property	LightGCN	H-GCN
Nodes	Customers + merchants (bipartite graph)	Merchants only (MCC tree)
Edges	Customer–merchant transactions	Parent–child hierarchy + brand co-visitation
Space	$\mathbb{R}^{64}$ Euclidean	$\mathbb{B}^8$ Poincaré Ball
Learning objective	“Who likes what”	“How are merchants structurally related”
Output	Customer embedding 64D (direct)	Merchant embeddings → per-customer aggregation (indirect)

2-Stage Pipeline: Stage 1 (offline) graph training → save embedding Parquet. Stage 2 (online) lookup + lightweight MLP adaptation. No graph propagation at inference — VRAM-friendly on a single GPU.

Other Features (Base Demographics, Product Holdings, Transaction Behavior)

Base Features 238D + 91D + 84D 413D / 644D

Base Features (238D)

The foundational feature block of traditional financial ML.

Subgroup	Dim.	Description
RFM (Recency, Frequency, Monetary)	34D	Multi-dimensional decomposition of transaction recency, frequency, and amount
Category	64D	Spending share/amount/frequency per MCC category
Transaction Stats	76D	Transaction statistics: mean, median, std. dev., skewness, kurtosis, etc.
Product Diversity	12D	Product holding diversity, cross-holding patterns
Channel Behavior	18D	Online/offline/mobile channel ratios
Temporal Pattern	22D	Transaction distribution by day-of-week and time-of-day
Demographics	12D	Age, gender, region, and other demographic attributes

Multi-Source Features (91D)

Subgroup	Dim.	Description
Deposit	25D	Deposit balance, inflow/outflow patterns, interest rate sensitivity
Credit	20D	Credit rating, utilization ratio, delinquency history
Investment	15D	Investment product holdings, risk preference
Digital Engagement	31D	App logs, login frequency, feature usage

Extended-Source Features (84D)

Subgroup	Dim.	Description
Insurance	20D	Insurance product holdings, subscription history
Refund/Cancellation	15D	Refund frequency, cancellation patterns
Consultation/STT	25D	Consultation history, speech recognition analysis

Subgroup	Dim.	Description
External Signals	24D	Signals from external data integrations

Model-Derived Features (27D)

Subgroup	Dim.	Description
HMM Summary	5D	Compressed representation of Triple-Mode 48D (dominant state, duration, stability, entropy, change rate)
Bandit/MAB	4D	Exploration-exploitation balance indicators from Multi-Armed Bandit
LNN Statistics	18D	Hand-crafted statistical features: moving average, volatility, autocorrelation, etc.

Merchant Hierarchy Features (27D)

Coordinates and embeddings reflecting the MCC hierarchy (Root $\rightarrow$ L1 $\rightarrow$ L2 $\rightarrow$ Brand). Includes MCC Poincaré embeddings extracted from H-GCN (Phase 0 v3/v4; was 21D stats-only prior to v3) and brand similarity based on co-visitation.

3-Stage Normalization Pipeline

Preprocessing Power-law -> Scaler -> Raw Copy 644D + 90D

Design Principles

The Normalization Pipeline consists of three stages. Its core objective is to **preserve the original magnitude information of power-law-distributed features** while transforming them to a scale suitable for model training.

Stage 1: Power-Law Detection + log1p Copy Generation

Power-Law Detection Criteria:

1. Candidate selection based on skewness and kurtosis
2. Power-law determination via R^2 of log-log regression

If power-law is confirmed: generate a `log1p` copy of the column for separate preservation

$$x_{\text{raw}} = \log(1 + x)$$

Power-law distributions (transaction amounts, balances, etc.) compress most values near zero when StandardScaler is applied due to outliers, resulting in information loss. The log1p transform mitigates this problem.

Stage 2: StandardScaler (TRAIN fit only)

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

Applied to: continuous columns only (excluding binary columns) **Core rule:** the Scaler must be fit exclusively on the TRAIN split. Val/test sets are only transformed using the scaler fit on train.

Data Leakage Prevention

Fitting the scaler on val/test data allows future information to leak into training, inflating estimated performance beyond its true value. This is the most common leakage pattern; LeakageValidator must verify this before training begins.

Stage 3: Preservation of Raw Power-Law Copy

The `_log` copies of power-law columns (generated in Stage 1) are preserved **without scaling**, retaining their original raw magnitude.

Why Preserve the Raw Copy?

StandardScaler redistributes information but **destroys absolute magnitude information**. For example, after z-score transformation, the difference between “monthly spend of 1,000,000 KRW” and “monthly spend of 10,000,000 KRW” is reduced to a relative position. However, for certain tasks (e.g., LTV prediction, spending_bucket classification), the absolute amount is critical. The 90D raw power-law copy provides this information directly to the model.

Final Tensor Composition

Segment	Dim.	Processing
Normalized continuous	554D	Stage 2 StandardScaler applied
Binary (pass-through)	90D	No scaling (0/1 preserved)
Raw power-law copy	90D	Stage 1 log1p only; Stage 2 not applied
Total	734D	644D normalized + 90D raw

Normalization Pipeline Flow Summary

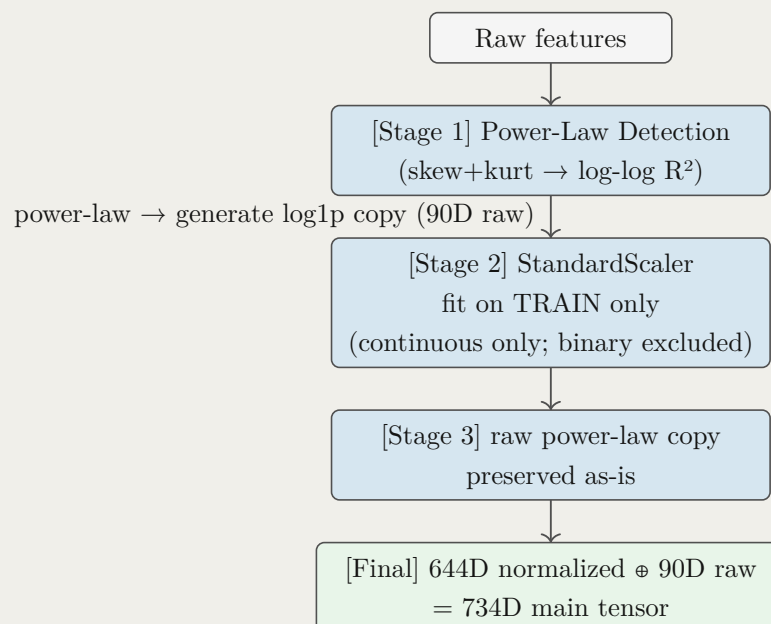


Figure 1: 3-stage normalization pipeline: power-law detection → StandardScaler → raw copy preservation.

Appendix: Design vs. Implementation Dimension Mapping

Note

This Appendix summarizes the dimensional differences between the full-bank design (734D) and the current Santander benchmark implementation (1211D input total, 17 feature groups). Implementation dimensions can be verified in `outputs/phase0/feature_schema.json`.

Feature Groups	Design (734D)	Implementation (1205D)	Notes
TDA	70D	32D	tda_global 16D + tda_local 16D
HMM	48D + 5D (separate)	25D	main tensor only
Base (Profile, etc.)	238D	53D	demographics 11D + product_holdings 24D + derived_temporal 4D + txn_behavior 14D
Graph	unspecified	100D	graph_collaborative 66D + product_hierarchy 34D
Merchant / Hierarchy	27D	27D	MCC Poincaré embeddings (merchant_hierarchy)
GMM + Model	22D	49D	gmm_clustering 22D + model_derived 27D
Lag Tensor (NEW)	—	800D	txn_lag_tensor: $K=200 \times 4$ features (lag_extractor)
Rolling Stats (NEW)	—	20D	txn_rolling_stats: 4 windows $\times$ 5 metrics (rolling_stats_extractor)
Multi-hot (NEW)	—	55D	nba_label_multihot 24D + mcc_top30_multihot 31D (topn_multihot_extractor)
Mamba	—	50D	mamba_temporal (GPU-precomputed, cached parquet)
Others (Economics, SIR, etc.)	335D	—	not implemented in Santander benchmark
Total	734D	1211D (17 groups)	17 feature groups

Further Reading

- **Economics generators:** Friedman 1957 (*Consumption Function*); Hall 1978 (PIH); Hodrick & Prescott 1997; Kalman 1960; Kahneman & Tversky 1979 (Prospect Theory).
- **Multidisciplinary:** Arrhenius 1889 (reaction rates); Kermack & McKendrick 1927 (SIR); Cohen & Felson 1979 (Routine Activity); Barabasi 2005 (burst heavy tails); Shannon 1948; Kullback & Leibler 1951.
- **TDA / PersLay:** Carlsson 2009; Edelsbrunner et al. 2002; Cohen-Steiner et al. 2007; Carriere et al. AISTATS 2020 (PersLay).
- **HMM:** Baum & Petrie 1966; Rabiner 1989; Dempster et al. 1977 (EM).
- **GMM:** Pearson 1894; Schwarz 1978 (BIC); Bishop 2006 (*PRML* Ch. 9).
- **Time series / Mamba:** Gu & Dao 2023 (Mamba); Cooley & Tukey 1965 (FFT).
- **Graph:** He et al. SIGIR 2020 (LightGCN); Chami et al. NeurIPS 2019 (HGNCN); Nickel & Kiela NeurIPS 2017 (Poincare); Rendle et al. UAI 2009 (BPR).